

Robert Wilms

Curriculum Vitae

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Research Interests

Arithmetic geometry, Arakelov geometry, Diophantine geometry, rational points, heights, equidistribution, Bogomolov-type results, algebraic curves, abelian varieties, moduli spaces, Riemann surfaces, (arithmetic) Newton–Okounkov bodies

Employment

- Since 2025 **Postdoctoral Researcher**, *University of Caen Normandy*, France.
Principal Investigator of the ERDF project **MERCI** (Mesure et recherche de la complexité par intersection), funded by the Région Normandie.
- 2023 – 2025 **Postdoctoral Researcher**, *University of Caen Normandy*, France.
Research group of Jérôme Poineau
- 2020 – 2023 **Postdoctoral Researcher**, *University of Basel*, Switzerland.
Research group of Philipp Habegger
- 2016 – 2019 **Postdoctoral Researcher**, *University of Mainz*, Germany.
Research group of Ariyan Javanpeykar

Education

- Since 2024 **Master studies of quantum technologies**, *University of Kaiserslautern-Landau*, Germany, Part-time distance studies.
- 2011 – 2016 **PhD studies of mathematics**, *University of Bonn / Max Planck Institute of Mathematics*, Germany.
Advisor: Gerd Faltings, defended 21.04.2016 (magna cum laude)
Thesis: The delta invariant in Arakelov geometry
- 2010 – 2011 **Master studies of mathematics**, *Ruhr University Bochum*, Germany.
Advisor: Markus Szymik
Thesis: Deligne–Tate Morphismen
- 2008 – 2010 **Bachelor studies of mathematics**, *Ruhr University Bochum*, Germany.
- 2007 – 2008 **Bachelor studies of mathematics**, *Technical University of Kaiserslautern*, Germany, Two-semester distance studies during my civilian service at the hospital.

Grants, Awards, and Scholarships

- 2025 **FEDER LABEL 2025**, *Région Normandie / ERDF*, Research Grant
Project: **MERCI** (Mesure et recherche de la complexité par intersection).
Budget: 110,000 €. Duration: 10/2025 – 03/2027.
- 2024 **Seal of Excellence: MSCA-IF High Quality Project**, *European Commission*, Certificate awarded for my project proposal “On Equidistribution in Arakelov theory”.
- 2011 – 2015 **International Max Planck Research School (IMPRS) PhD Scholarship**, *Max Planck Society*.
- 2012 **Studierenden Preis 2012**, *Ruhr University Bochum*, Student Award for the best master thesis in mathematics.

Research Visits

- Jan – Apr 2018 **Institute des Hautes Études Scientifiques**, *Invited researcher*, Bures-sur-Yvette, France.
- Feb – Apr 2010 **Max Planck Institute of Mathematics**, *Internship as a bachelor student*, Bonn, Germany.

Publications

Peer-reviewed, published articles

- *On the additivity of Newton-Okounkov bodies*. Beitr. Algebra Geom. (2025). <https://doi.org/10.1007/s13366-025-00807-9>, 16 pages.
- (with N. Looper and J. Silverman) *A uniform quantitative Manin–Mumford theorem for curves over function fields*. J. Reine Angew. Math. **828** (2025), 127–147.
- *Some differential equations for the Riemann θ -function on Jacobians*. SIGMA Symmetry Integrability Geom. Methods Appl. **20** (2024), Paper No. 060, 11 pp.
- (with N. M. Mavraki and H. Schmidt) *Height coincidences in products of the projective line*. Math. Z. **304** (2023), no. 2, Paper No. 26, 9 pp.
- *On arithmetic intersection numbers on self-products of curves*. J. Algebraic Geom. **31** (2022), no. 3, 397–424.
- *New explicit formulas for Faltings’ delta-invariant*. Invent. Math. **209** (2017), no. 2, 481–539.
- (with G. Gallot and P. Moree) *The family of ternary cyclotomic polynomials with one free prime*. Involve **4** (2011), no. 4, 317–341.

Preprints

- *Intersection numbers as mixed volumes of Newton–Okounkov bodies.* arXiv.org/abs/2502.18441, 12 pages.
- *An alternative proof of the Faltings–Elkies bound.* arXiv.org/abs/2310.03648, 11 pages.
- *On the Irreducibility and Distribution of Arithmetic Divisors.* arxiv.org/abs/2211.03766, 70 pages.
- *Degeneration of Riemann theta functions and of the Zhang–Kawazumi invariant with applications to a uniform Bogomolov conjecture.* arXiv.org/abs/2101.04024, 23 pages.

Other publications

- *On the distribution of algebraic numbers.* In: Diophantische Approximationen (hybrid meeting), MFO report no. **21/2022** (2022), p. 1107–1109.
- *On the delta invariant in Arakelov geometry.* In: Arithmetic Geometry, MFO report no. **38/2016** (2016), p. 2175–2177.
- *The delta invariant in Arakelov geometry.* PhD thesis, University of Bonn (2016), urn:nbn:de:hbz:5n-43447.

Invited Talks

- Nov 2025 **Arakelov theory on Noetherian schemes, Diophantine approximation and transcendence**, CIRM, Luminy.
- Jul 2025 **On the quantitative Manin–Mumford conjecture, Unlikely Intersections in arithmetic and dynamical systems**, University of Warwick.
- Jun 2025 **A purely algebraic intersection theory, Modular in Bielefeld**, University of Bielefeld.
- May 2025 **Sur la distribution des nombres algébriques et des généralisations en géométrie algébrique, 17e journée de la Fédération Normandie-Mathématiques**, Université de Rouen-Normandie.
- Apr 2025 **An intersection theory for norms on rings, Séminaire de géométrie algébrique de Jussieu**, IMJ-PRG.
- Feb 2025 **An algebraic approach to (arithmetic) intersection theory, Number Theory Seminar**, University of Warwick.
- Oct 2024 **On the quantitative Manin–Mumford conjecture, Oberseminar Diophantische Geometrie**, University of Göttingen.
- Sep 2024 **Valuative compactifications of complex varieties with applications to moduli spaces of curves, Paris Arithmetic Homotopic Galois Theory Days 2024**, Institut de mathématiques de Jussieu – Paris Rive Gauche.
- May 2024 **A new construction of arithmetic Okounkov bodies, Special day on Okounkov bodies**, École polytechnique.
- Apr 2024 **On irreducibility in Arakelov theory, Séminaire de géométrie**, École polytechnique.

- Mar 2024 **On equidistribution in Arakelov theory**, *Séminaire de Théorie des nombres*, Grenoble Alpes University.
- Nov 2023 **On the quantitative Manin–Mumford conjecture**, *Séminaire de Théorie des Nombres*, University of Caen Normandy.
- Sep 2023 **On equidistribution in Arakelov theory**, *4th Westlake AGNT seminar*, Westlake University.
- Aug 2023 **On Okounkov bodies in Arakelov theory**, *Masterclass: Arakelov geometry on Shimura varieties*, University of Copenhagen.
- Feb 2023 **On the irreducibility and distribution of arithmetic divisors**, *Séminaire de Théorie des Nombres*, University of Caen Normandy.
- Jan 2023 **On equidistribution in Arakelov theory**, *Number Theory Web Seminar*, Zoom.
- Jan 2023 **On equidistribution in Arakelov theory**, *Number Theory Seminar*, University of Copenhagen.
- Jan 2023 **On the irreducibility of arithmetic cycles**, *Colloquium in Honor of the Retirement of Professor Gerd Faltings*, Max Planck Institute for Mathematics Bonn.
- Sep 2022 **A quantitative Bogomolov-type result for curves over function fields**, *BIRS Workshop "Specialisation and Effectiveness in Number Theory"*, Banff International Research Station.
- Jun 2022 **On the irreducibility of arithmetic divisors**, *Workshop "Alpbach 2022"*, Alpbach
- Apr 2022 **On the distribution of algebraic numbers**, *MFO Workshop Diophantine Approximationen*, MFO Oberwolfach.
- Mar 2022 **On the distribution of algebraic numbers and its generalization in Arakelov geometry**, *Rhine Seminar on Transcendence Basel-Freiburg-Strasbourg*, University of Freiburg.
- Dec 2021 **On the asymptotics of Arakelov invariants**, *Séminaire "Fables Géométriques"*, University of Geneva.
- Oct 2021 **A quantitative Bogomolov-type result for function fields**, *Number Theory Seminar*, University of Copenhagen.
- Sep 2021 **A uniform Bogomolov type result for tautological cycles in Jacobians of curves in one-parameter families**, *Intercity seminar on Arakelov theory*, University of Regensburg.
- Nov 2017 **On Faltings' delta-invariant**, *Basel Number Theory Seminar*, University of Basel.
- Oct 2017 **Local descriptions of Arakelov invariants**, University of Frankfurt.
- Jun 2017 **On Faltings' delta invariant**, Technical University of Darmstadt.

- Dec 2016 **New formulas for Faltings' delta-invariant**, *Séminaire Théorie des Nombres*, University of Bordeaux.
- Sep 2016 **New formulas for Faltings' delta invariant**, *Intercity seminar on Arakelov theory*, University of Regensburg.
- Aug 2016 **On the delta invariant in Arakelov theory**, *Workshop on arithmetic geometry*, MFO Oberwolfach.
- Jul 2016 **On Faltings' delta invariant**, *Mittagsseminar*, University of Münster.
- Jan 2016 **New formulas for Faltings' delta invariant**, University of Mainz.

Supervision

- 2023 **Lena Redlich**, *Master thesis*, "Consequences of the abc-Conjecture in Arithmetic Dynamics".

Organization

- 2025 Co-organizer of the Conference "Arithmetic Geometry in Cabourg".
- 2022 – 2023 Co-organizer of the Basel Number Theory Seminar.
- Fall 2021 Co-organizer of a Zoom seminar on *Adelic line bundles over quasi-projective varieties*.
- WS 2016 Co-organizer of a seminar on *Derived Categories* at the University of Mainz.
- 2014 Co-organizer of the workshop *Kleine AG on Arakelov geometry and the Bogomolov conjecture*.
- 2014 Co-organizer of the workshop *Kleine AG on mod p local Langlands correspondence over $\mathbb{Q}_p$* .

Teaching Experience

Teaching experience as Lecturer

University of Basel

- Spring 2022 Einführung in die Topologie.

Max Planck Institute for Mathematics, Bonn

- WS 2014/15 Étale Cohomology (Lecture series for PhD students).

Teaching experience as a Teaching Assistant

University of Basel

- Spring 2023 Lineare Algebra II.
- Fall 2022 Komplexe Dynamik.
- Fall 2021 Algebraische Zahlentheorie.

Spring 2021 Lineare Algebra II.
Fall 2020 Zahlentheorie.
Spring 2020 Lineare Algebra II.

University of Mainz

WS 2019/20 Lineare Algebra.
WS 2018/19 Zahlentheorie.
SS 2018 Algebra II.
SS 2017 Algebra II.
WS 2016/17 Lineare Algebra II.
SS 2016 Computeralgebra.

Ruhr University of Bochum

2009 – 2011 Tutor for lectures in algebra, geometry and representation theory.

Selected attended Conferences

Nov 2025 **Diophantine approximation and transcendence**, *CIRM, Luminy*.
Jul 2025 **Unlikely Intersections in arithmetic and dynamical systems**, *University of Warwick*.
Jun 2025 **Modular in Bielefeld**, *University of Bielefeld*.
Jul 2024 **Conference on “Arithmetic Geometry” in Honour of Gerd Faltings’ 70th Birthday**, *MPIM, Bonn*.
Oct 2023 **Global invariants of arithmetic varieties**, *CIRM, Marseille*.
Aug 2023 **Masterclass: Arakelov geometry on Shimura varieties**, *University of Copenhagen*.
Jun 2023 **XXXIèmes Rencontres Arithmétiques de Caen: Théorie d’Arakelov et Géométrie Diophantienne**, *Île de Tatihou*.
Sep 2022 **School on Arithmetic Geometry**, *University of Essen*.
Sep 2022 **BIRS Workshop ”Specialisation and Effectiveness in Number Theory”**, *Banff International Research Station*.
Jul 2022 **Number Theory and Algebraic Geometry, ICM sectional workshop**, *ETH Zürich*.
Jun 2022 **Workshop ”Alpbach 2022”**, *Alpbach*.
Apr 2022 **Workshop Diophantine Approximationen**, *MFO Oberwolfach*.
Sep 2021 **10th Swiss-French workshop in Algebraic Geometry**, *Charmey*.
Sep 2021 **Intercity Seminar on Arakelov Theory**, *University of Regensburg*.
Jan 2020 **9th Swiss-French workshop in Algebraic Geometry**, *Charmey*.

- Jun 2019 **Geometry and Arithmetic of Algebraic Varieties, Final Conference of the SFB/TRR45 Bonn/Essen/Mainz**, *University of Bonn*.
- Mar 2019 **Workshop on prismatic cohomology**, *University of Essen*.
- Oct 2018 **Conference on Arithmetic Algebraic Geometry – On the occasion of Michael Rapoport’s 70th birthday**, *University of Bonn*.
- Jul 2018 **Berkovich Spaces: 30 Years**, *Institut Henri Poincaré, Paris*.
- Jun 2018 **Arithmetic and Algebraic Geometry – A conference in honor of Ofer Gabber on the occasion of his 60th birthday**, *Institute des Hautes Études Scientifiques*, Invited researcher, Bures-sur-Yvette.
- Apr 2018 **Arithmetic and Analysis – A conference on the occasion of Christopher Deninger’s 60th birthday**, *University of Münster*.
- Sep 2016 **Intercity Seminar on Arakelov Theory**, *University of Regensburg*.
- Aug 2016 **Workshop Arithmetic Geometry**, *MFO Oberwolfach*.
- Jun 2016 **Arakelov theory and automorphic forms – A conference at the occasion of Jürg Kramer’s 60th birthday**, *Humboldt University of Berlin*.
- Oct 2015 **Panorama of Mathematics**, *University of Bonn*.
- Jun 2014 **Conference on Arithmetic Algebraic Geometry on the occasion of Gerd Faltings’ 60th birthday**, *University of Bonn*.
- Jan – Apr 2014 **Junior Hausdorff Trimester Program: Algebraic geometry**, *Hausdorff Research Institute for Mathematics, Bonn*.
- Jan – Apr 2013 **Hausdorff Trimester Program: Arithmetic and geometry**, *Hausdorff Research Institute for Mathematics, Bonn*.
- Aug 2012 **Workshop Arithmetic Geometry**, *MFO Oberwolfach*.
- Dec 2011 **Opening of the Floer Center of Geometry**, *Ruhr University Bochum*.
- Oct 2011 **DMV Jahrestagung**, *University of Cologne*.

Further Activities

- Referee Bulletin des sciences mathématiques, Duke Mathematical Journal, Expositiones Mathematicae, International Mathematics Research Notices, Journal für die Reine und Angewandte Mathematik, Journal of Algebraic Combinatorics, Journal of Algebraic Geometry, Journal of Differential Geometry, Proceedings of the American Mathematical Society, Pure and Applied Mathematics Quarterly, Transactions of the American Mathematical Society.
- Reviewer MathSciNet, zbMATH.
- Programming skills Python, Latex, Sage, Java, Basic, Fortran.
- Language skills German (native), English (fluent), French (intermediate).